

Adaptive Segmentation for Robust Texture-Aware Voxelization under Varying Imaging Conditions

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1. Motivation

Voxelization pipelines often rely on fixed segmentation parameters and a single color quantization strategy. In practice, however, input views or texture maps may vary significantly in brightness, contrast, and noise level, which can cause unstable boundaries, missing thin structures, and poor palette quality. This project proposes an adaptive image-processing front end for voxelization, using segmentation and image statistics to guide both geometry conversion and color mapping.

2. Objective

The goal of this project is to design an adaptive segmentation-guided voxelization pipeline for textured 3D models. Instead of using fixed parameters, the system will:

- analyze 2D texture maps or rendered views of the input model
- estimate visual conditions such as brightness, contrast, noise level, and edge density.
- automatically tune segmentation parameters and select voxelization and color mapping methods

3. Method

3.1 Adaptive Segmentation

First, we will implement several segmentation baselines, including thresholding, k-means, and mean-shift, with graph-based segmentation as an optional extension. The key idea is not to keep their parameters fixed. For example, darker images may require higher threshold sensitivity, low-contrast inputs may need contrast enhancement before segmentation, and noisy images may benefit from stronger smoothing or a larger mean-shift bandwidth.

3.2 Integration with Voxelization

Second, the segmentation results and image statistics will guide voxelization. We plan to compare triangle rasterization, subdivision-based voxelization, and ray-based voxelization. Triangle rasterization is expected to be efficient for clean surfaces, subdivision may better preserve thin structures and detailed boundaries, and ray-based methods may better handle ambiguous occupancy near the surface. The adaptive system will either choose one method globally per model or use segmentation complexity to guide method selection.

3.3 Color Mapping

We further analyze how segmentation influences color representation.. We will study palette generation and color mapping using median cut, k-means, and mean-shift. Median cut will serve as the baseline, while k-means will be used for compact fixed-size palettes and mean-shift will be tested for cases where the number of dominant colors is unknown. The selection of color mapping strategy will depend on region complexity and color distribution in the segmented image.

4. Dataset and Experimental Setup

We plan to construct a controlled dataset of textured 3D models generated using modern 3D generation tools, enabling systematic evaluation under varying imaging conditions. We will compare a fixed-parameter baseline against the proposed adaptive pipeline. Ablation studies will include: fixed segmentation plus fixed voxelization/color mapping, adaptive segmentation only, adaptive color mapping only, and the full adaptive system.

5. Evaluation

The final voxel results will be rendered from the same viewpoints as the original models and compared using image-based metrics such as SSIM/PSNR, silhouette IoU or boundary preservation, and color error in RGB or Lab space. We will also report runtime, voxel count, and palette size. Since the Voxelt pipeline also targets brick-style outputs, we may additionally report practical buildability-related statistics such as filled voxel count, brick count, and unique color count.

6. Expected Contribution

This project connects classical image segmentation with a practical mesh-to-voxel pipeline. The expected contribution is an adaptive framework that improves robustness under varying lighting and noise conditions, while preserving both boundary detail and color fidelity better than fixed-parameter baselines.